Semidefinite Programming (SDP) & Sum of Squares (SOS) (Lec13.pdf)

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Todays topics & Logistics

- **S**emi**D**efinite **P**rogramming (SDP)
- Application on nonconvex polynomial optimization
- Nonnegative polynomials
- Sum of Squares (SOS) polynomials
- SOS & SDP
- SOS ≠ …---…

- **HW6** → due 4/10 @ 11:59PM
- **HW7 & HW8** → released 4/11
  - Due on 4/28 and May 5 @ 11:59PM
- **Preparations for Final Exam……**
  - Lecture notes & slides & HWs
  - 24h-48h take home…
  - Past final exams for practice

The symbols $\geq$ and $\succcurlyeq$

- **Remark**: The symbols $\geq$ and $\succcurlyeq$ have different meanings.
  - Consider the matrix $X \in S^{n \times n}$
  - $\geq$ means <u>elementwise</u> nonnegativity of the elements of X
    - $X_{ij} \geq 0, \forall i, j$ $\qquad X = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix} \geq 0$, but $X \not\succcurlyeq 0$
  - $\succcurlyeq$ means the matrix X is PSD

$$Y = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \succcurlyeq 0, \quad \text{but } Y \not\geq 0$$

SDP – refresher

- **Definition**: An SDP is an optimization problem of the form:

$$\begin{array}{ll} min & Tr(CX) \\ s.t. & Tr(A_i X) = b_i, i = 1, \dots, m \\ & X \succcurlyeq 0 \end{array}$$

  - **Data**: $C \in S^{n \times n}$, $A_i \in S^{n \times n}$, $b_i \in R$, $i = 1, \dots, m$
  - $S^{n \times n}$ is the set of Symmetric $n \times n$ real matrices: $S^{n \times n} \subset R^{n \times n}$
  - $Tr(H)$ is the **Trace** of the square matrix $H \in S^{n \times n}$:
    - $Tr(H) = \sum_{i=1}^{n} h_{ii}$ → the **sum of the diagonal elements**

SDP – refresher

- Why do we need the TRACE notation??

- It is a convenient way to denote a "linear function" in terms of the elements of the matrix X

$$Tr(CX) = \sum_{i,j=1}^{n} c_{ij} x_{ij}$$

Example 1: $H = CX = \begin{pmatrix} c_{11} & c_{12} \\ c_{12} & c_{22} \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} \\ x_{12} & x_{22} \end{pmatrix} = \begin{pmatrix} c_{11}x_{11} + c_{12}x_{12} & c_{11}x_{12} + c_{12}x_{22} \\ c_{12}x_{11} + c_{22}x_{12} & c_{12}x_{12} + c_{22}x_{22} \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix}$

$$Tr(H) = Tr(CX) = (c_{11}x_{11} + c_{12}x_{12}) + (c_{12}x_{12} + c_{22}x_{22}) = h_{11} + h_{22}$$

SDP – refresher

- **Example 2:** consider the inequality: $4X_{11} - 6X_{12} + 5X_{23} \leq 7$
  - How can we write it in the form: $Tr(AX) \leq b$??
  - We need to find: b, A, X
  - $b = 7$, $A = \begin{pmatrix} 4 & -3 & 0 \\ -3 & 0 & 5/2 \\ 0 & 5/2 & 0 \end{pmatrix}$, $X = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{12} & x_{22} & x_{23} \\ x_{13} & x_{23} & x_{22} \end{pmatrix}$

Nonconvex Polynomial Optimization

- Consider the **polynomial function** $p: R^n \to R$

- And the unconstrained (nonconvex) polynomial optimization problem

$$\boxed{\begin{array}{ll} min & p(x) \\ s.t. & x \in R^n \end{array}}$$

We are looking for the **Global minimum**
(it's $\boldsymbol{NP}$**-hard** problem;
more details next week)

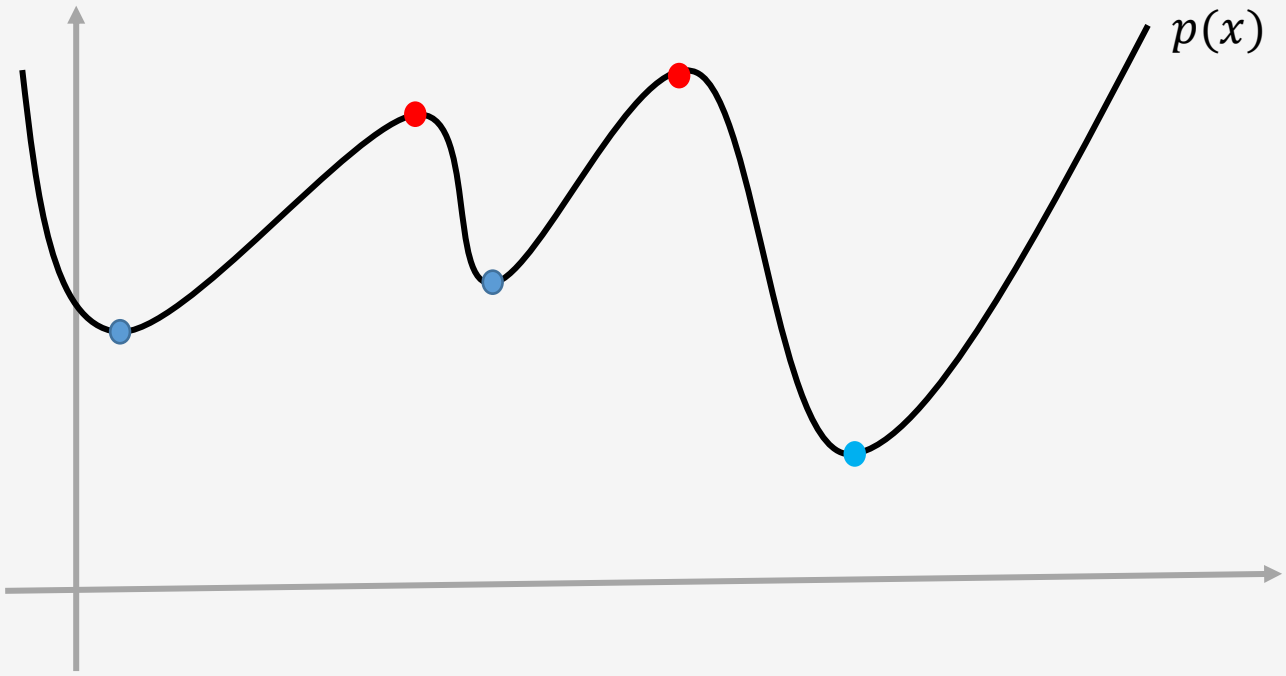
- All the optimization algorithms we have learned so far (e.g., steepest descent, Newton method) can be used to solve this problem.

- BUT they converge to a **Local minimum** $\boldsymbol{x}^*$.

  - We can use the optimality conditions for deciding whether $\boldsymbol{x}^*$ is local minimum

Nonconvex Polynomial Optimization

- General plot of a nonconvex polynomial



NP-hard

$$\min_{x} \; p(x)$$
$$\text{s.t. } x \in R^n$$

We want to find the Global minimum.

Where should I start??

Nonconvex Polynomial Optimization

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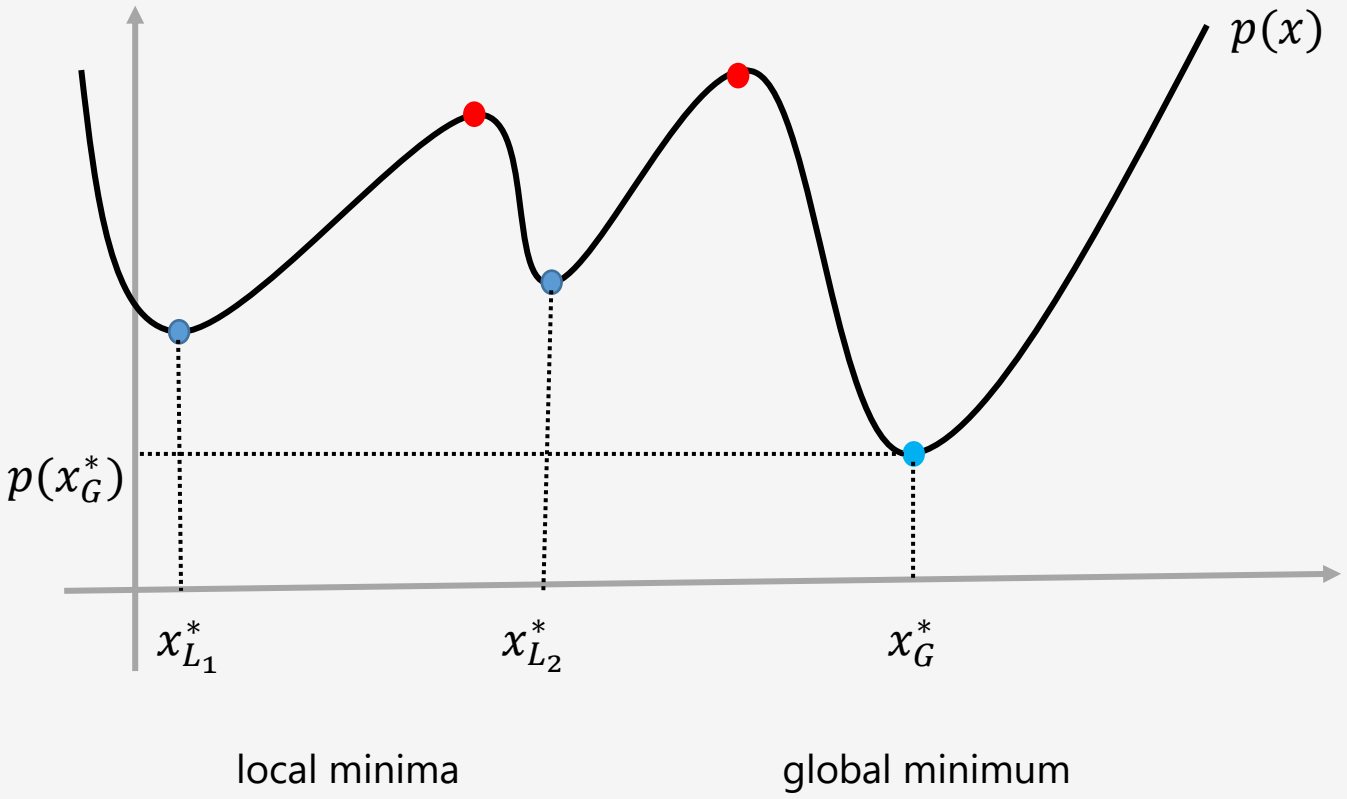

***NP*-hard**

$$\min_x \; p(x)$$
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We want to find the Global minimum.

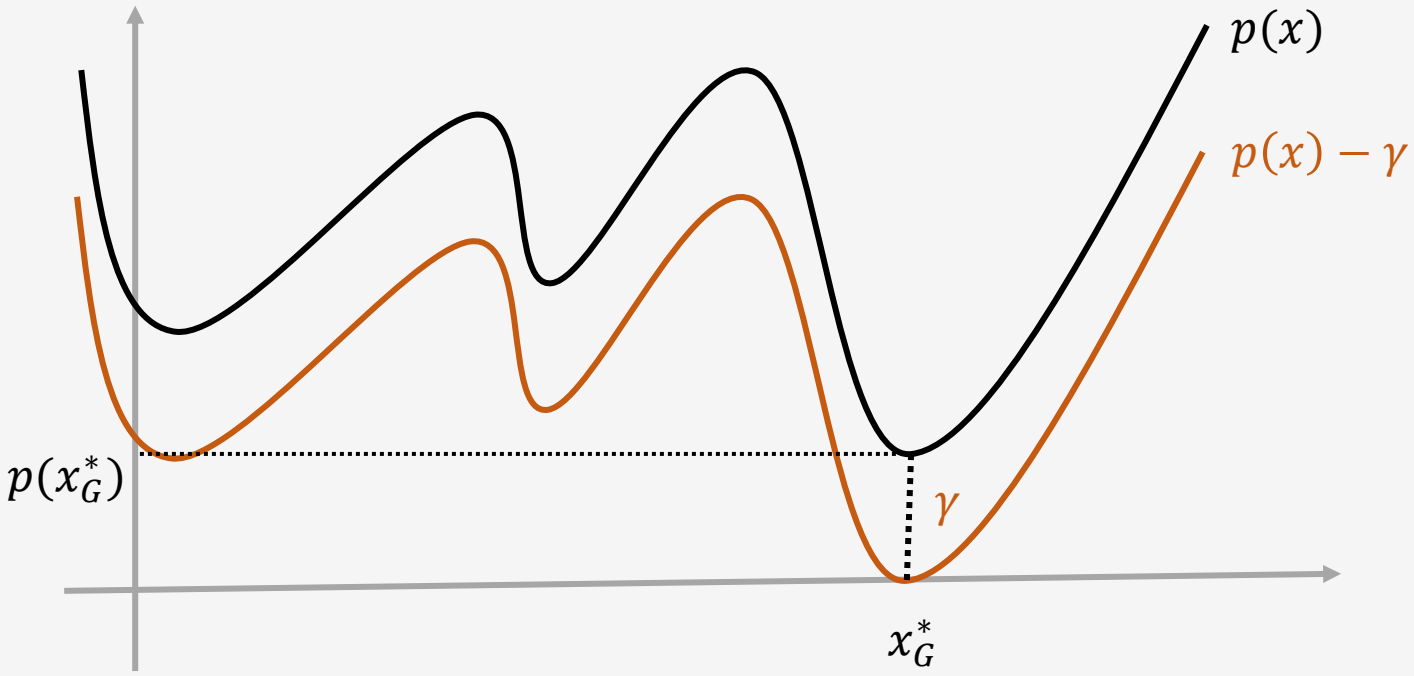
Where should I start??

Regions of attraction for each local optimum??

Do you see the analogy with **Dynamical Systems**, Stability & Lyapunov functions ??

Nonconvex Polynomial Optimization – NEW perspective

- General plot of a nonconvex polynomial



***NP*-hard**

$$\min_{x} \; p(x)$$
$$s.t. \; x \in R^n$$

$$\Updownarrow$$

$$\max_{\gamma} \; \gamma$$
$$s.t. \quad p(x) - \gamma \geq 0, \forall x \in R^n$$
$$\gamma \in R$$

We are looking for the **maximum quantity** γ **we can subtract** from $\boldsymbol{p(x)}$ and still have $\boldsymbol{p(x) - \gamma \geq 0}$ everywhere!

Question: What sign γ would have in this case ??

Nonconvex Polynomial Optimization – NEW perspective

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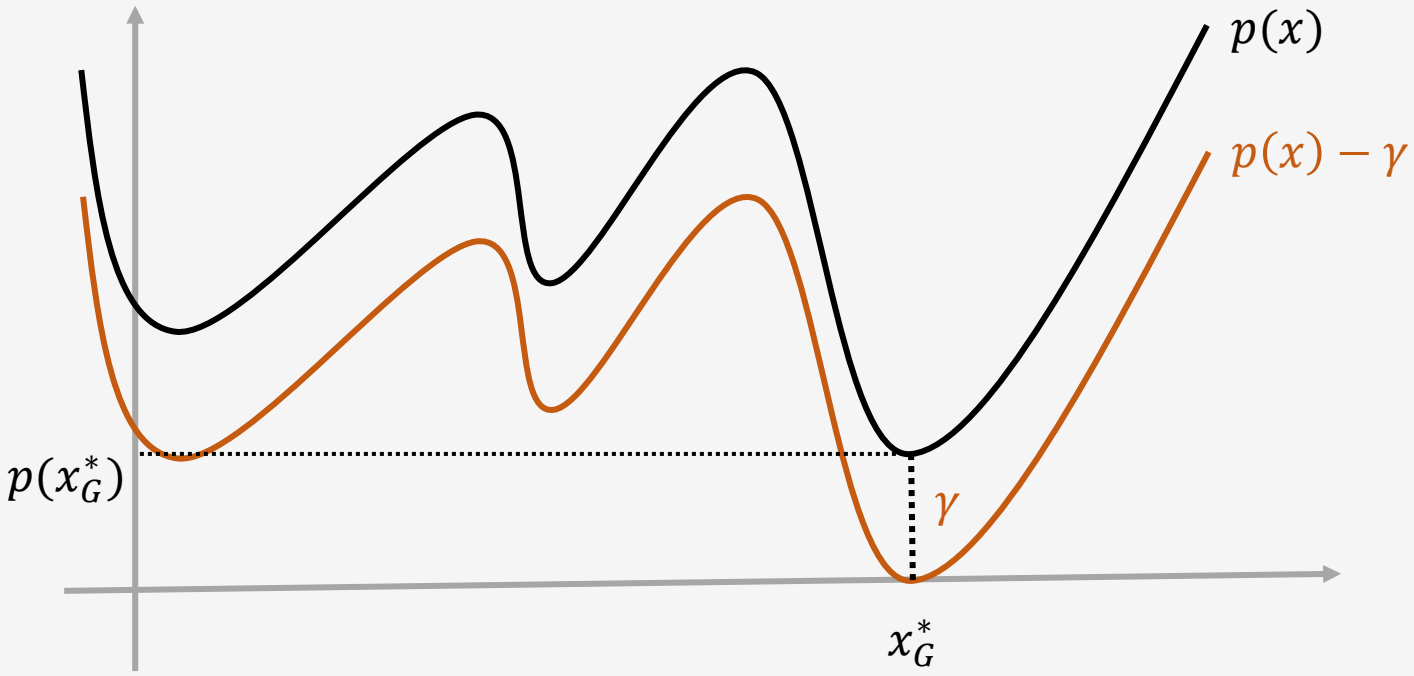

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Question: What sign γ would have in this case ?? → **Answer**: $\gamma \geq 0$

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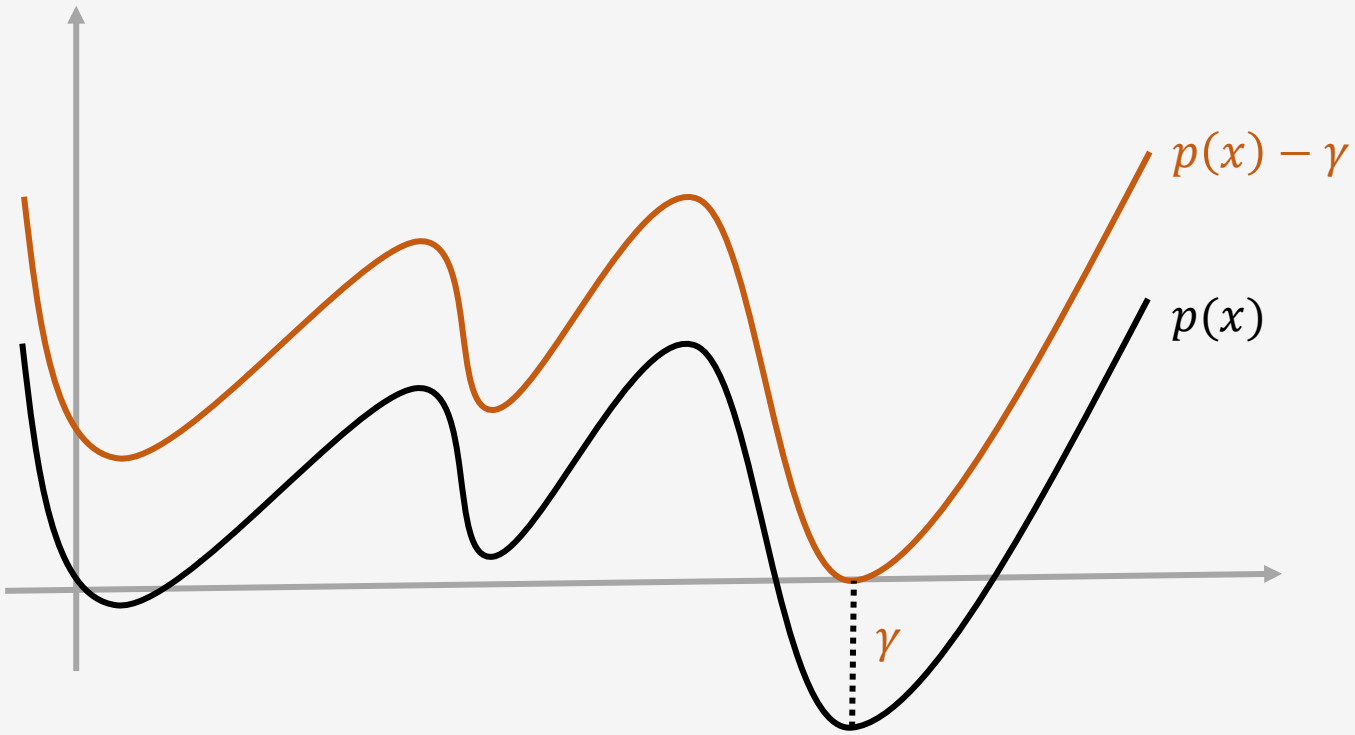

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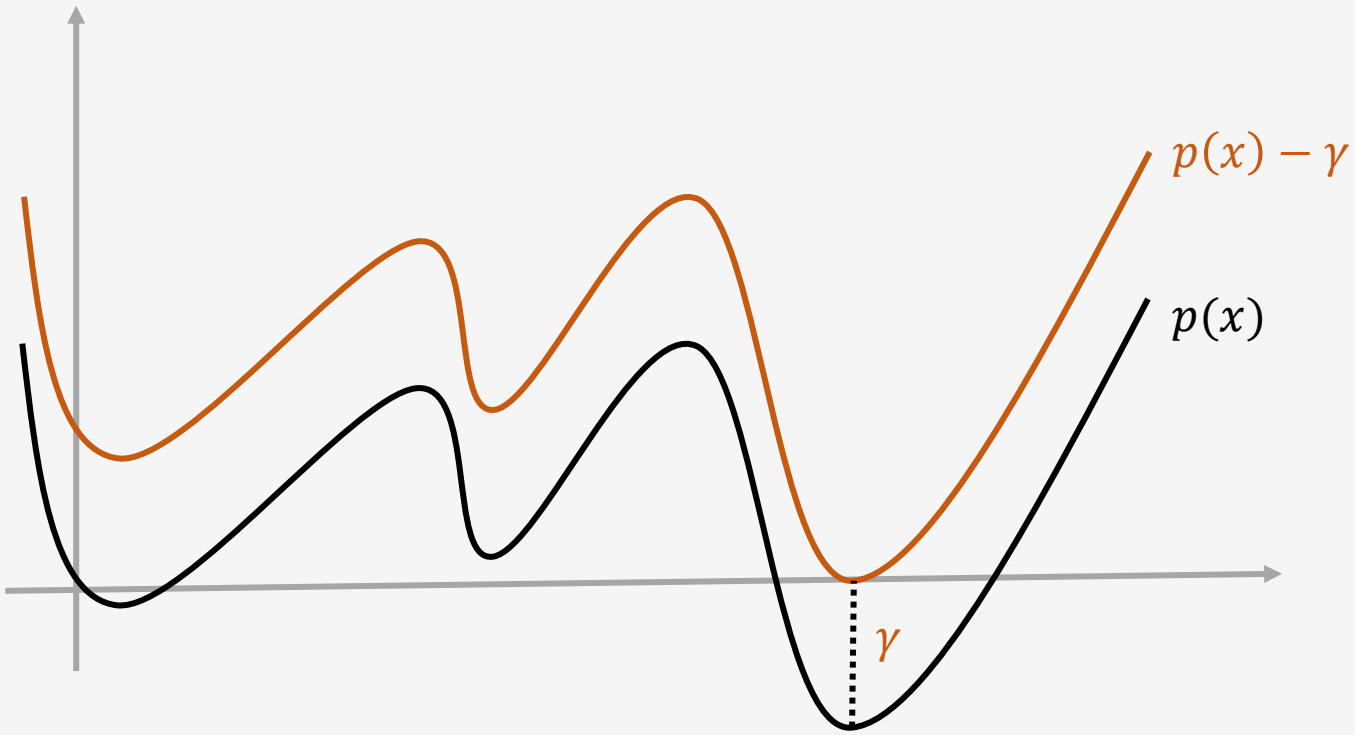

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Question: What sign γ would have in this case ?? → **Answer**: $\gamma \leq 0$

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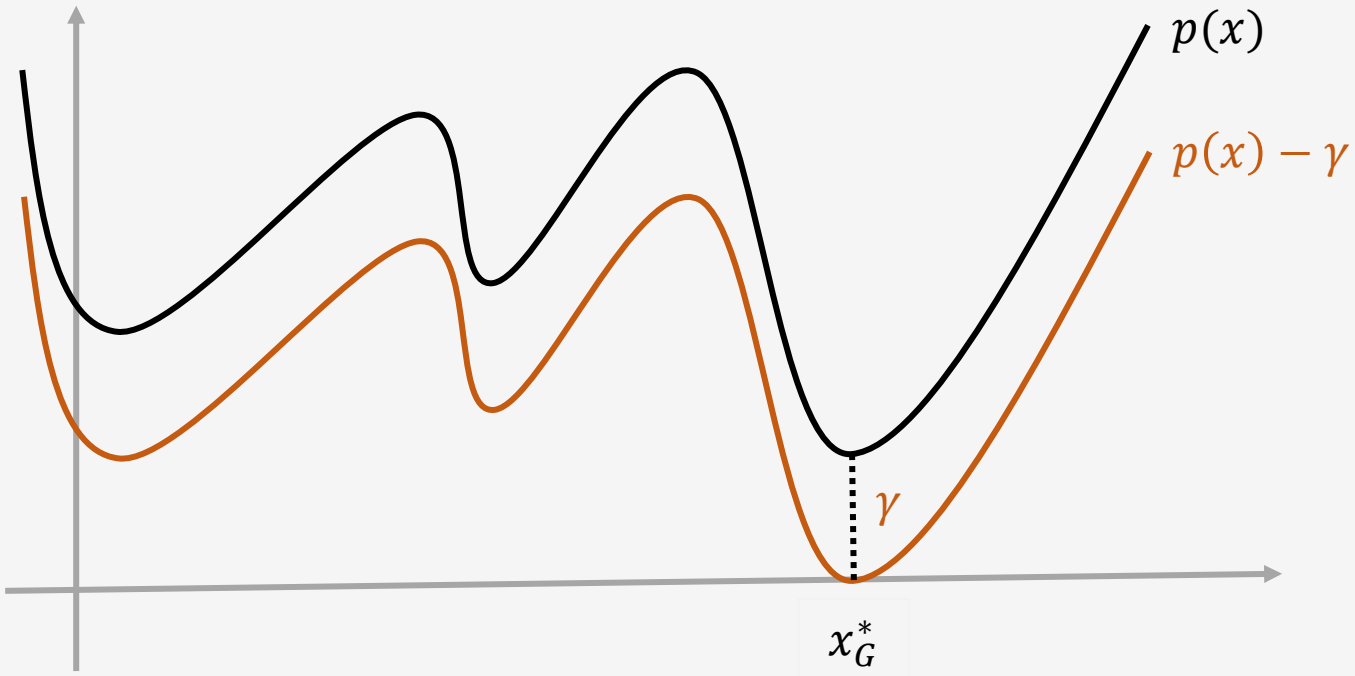

***NP*-hard**

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ \text{s.t. } x \in R^n \end{array}\right\} \quad (1)$$

$$\Updownarrow \; (=)$$

$$\left.\begin{array}{ll} \max\limits_{\gamma} & \gamma \\ \text{s.t.} & p(x) - \gamma \geq 0, \forall x \in R^n \\ & \gamma \in R \end{array}\right\} \quad (2)$$

Advantage of new formulation → It has only one decision variable $\gamma \in R$ (note: x are fixed)

Disadvantage of new formulation → the constraint $p(x) - \gamma \geq 0, \forall x \in R^n$ is difficult to satisfy → it introduces an ∞ number of constraints, because it needs to be satisfied $\forall x \in R^n$.

Nonnegative polynomials

- **Definition**: a polynomial $\boldsymbol{p}: R^n \to R$ is called nonnegative polynomial **iff** $p(x) \geq 0, \forall x \in R^n$
- So, if we find out how to characterize when a polynomial is nonnegative, we could handle the difficult constraint: $p(x) - \gamma \geq 0, \forall x \in R^n$
- **Question**: can you tell me some nonnegative polynomials ??

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- **Question**: can you tell me some nonnegative polynomials ??
  - **Examples**: $x^2, \quad x^4, \quad x^2 + x^4, \quad x^2 + x^6, \ldots$
  - What characteristic do they have ??

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    - $(x)^2,\ \ (x^2)^2, (x)^2 + (x^2)^2,\ \ (x)^2 + (x^3)^2, \ldots$

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  - What characteristic do they have ??
    - $(x)^2,\ \ (x^2)^2, (x)^2 + (x^2)^2,\ \ (x)^2 + (x^3)^2, \ldots$ → **Sums of Squares (SOS)**

Sum of Squares – SOS

- **Definition**: A polynomial $p: R^n \to R$ is a sum of squares (SOS) iff

$$p(x) = \sum_{i=1}^{m} q_i^2(x)$$

for some other m polynomials $q_i: R^n \to R, i = 1,2, \dots, m.$

$$q_i(x) = a_i x_1^{k_{i1}} x_2^{k_{i2}} \dots x_n^{k_{in}}$$

$$a_i \in R, \qquad k_{ij} \in Z_+$$

- **Some Observations**:

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- **Some Observations**:
  - $p(x)$ is SOS $\Rightarrow$ $p(x)$ is nonnegative polynomial
  - $p(x)$ is nonnegative polynomial $\not\Rightarrow$ $p(x)$ is SOS (in general…)

Sum of Squares – SOS

- Good news: we can impose the constraint that a nonnegative polynomial is SOS, using **SDP**.

- Observations about problems (2) and (3):

$$\left.\begin{array}{l}\min\limits_{x}\ p(x)\\ s.t.\ x \in R^n\end{array}\right\}\ (1)$$

$$\Updownarrow\ (=)$$

$$\left.\begin{array}{ll}\max\limits_{\gamma} & \gamma\\ s.t. & \underbrace{\boxed{\begin{array}{c}p(x)-\gamma \geq 0, \forall x \in R^n\\ \gamma \in R\end{array}}}_{\Omega_2}\end{array}\right\}\ (2)$$

$$\textbf{??}\ [\leq \boldsymbol{or}\ = \boldsymbol{or}\ \geq]\ \textbf{??}$$

$$\left.\begin{array}{ll}\max\limits_{\gamma} & \gamma\\ s.t. & \underbrace{\boxed{\begin{array}{c}p(x)-\gamma\ is\ SOS\\ \gamma \in R\end{array}}}_{\Omega_3}\end{array}\right\}\ (3)$$

Sum of Squares – SOS

- Good news: we can impose the constraint that a nonnegative polynomial is SOS, using **SDP**.

- Observations about problems (2) and (3):

  - They have the same objective function

  - What is the relationship between Ω_2 and Ω_3 ??

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    - $p(x)$ is SOS $\Rightarrow$ $p(x)$ is nonnegative poly

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?? $[\leq$ ***or*** $=$ ***or*** $\geq]$ **??**

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  - What is the relationship between Ω_2 and Ω_3 ??
    - $p(x)$ is SOS $\Rightarrow$ $p(x)$ is nonnegative poly
    - Hence $\Omega_3 \subseteq \Omega_2$

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ s.t. \; x \in R^n \end{array}\right\} (1)$$

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$$\textbf{??} \; [\leq \textbf{or} \; = \textbf{or} \; \geq] \; \textbf{??}$$

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Sum of Squares – SOS

- Good news: we can impose the constraint that a nonnegative polynomial is SOS, using SDP.

- Observations about problems (2) and (3):
  - They have the same objective function
  - What is the relationship between Ω_2 and Ω_3 ??
    - $p(x)$ is SOS $\Rightarrow$ $p(x)$ is nonnegative poly
    - Hence $\Omega_3 \subseteq \Omega_2$ → Problem (3) provides a **LOWER BOUND** to problems (2) and (1). (i.e., problem (3) is a ***relaxation*** of problems (2) and (1))

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ s.t. \; x \in R^n \end{array}\right\} (1)$$

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IV Less than Or equal

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Sum of Squares – SOS

- Recall: $p(x) \geq 0, \forall x \not\Rightarrow p(x)$ is SOS
  - **Counterexamples are very difficult to construct.**
  - $p(x,y) = 1 + x^2y^4 + x^4y^2 - 3x^2y^2$ [Motzkin, 1967]
- **Special cases [proved by D. Hilbert, 1888]:**

1. If $n = 1$: $p(x) \geq 0, \forall x \in R \Rightarrow p(x)$ is SOS
2. If $\deg(p(x)) = 2$: $p(x) \geq 0, \forall x \in R^n \Rightarrow p(x)$ is SOS
3. If $\deg(p(x)) = 4 \,\&\, n = 2$: $p(x) \geq 0, \forall x \in R^2 \Rightarrow p(x)$ is SOS

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ s.t. \; x \in R^n \end{array}\right\} (1)$$

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SOS and SDP

- **<u>Theorem</u>**: A polynomial $p: R^n \to R$ of $\deg(p) = 2d$ is SOS $\Leftrightarrow$ there exists a matrix $Q \in S^{n\times n}$ and $Q \succcurlyeq 0$ such that $p(x) = z(x)^T Q z(x)$, where $z(x)$ is a vector of all monomials in x of degree $\leq d$. *(Recall that a **monomial** is a **polynomial of only one term**)*

Solving SDPs has $\boldsymbol{Polynomial\ time}$ **complexity** (we will discuss this next week)

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- What is the vector $z(x)$??

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- What is the vector $z(x)$??

  - **Example**: if $n = 2$ and $2d = 4$ the vector $z(x)$ is defined as:

$$z(x) = (1, x_1, x_2, x_1^2, x_2^2, x_1 x_2)^T \in R^{\boldsymbol{N=6}}$$

    - $\boldsymbol{N}$ is defined by: $\mathbf{N} = \binom{\boldsymbol{n}+\boldsymbol{d}}{\boldsymbol{d}} = \frac{(\boldsymbol{n}+\boldsymbol{d})!}{\boldsymbol{d}!\,\boldsymbol{n}!}$ and for $n = 2,\ 2d = 4$ we get $\boldsymbol{N} = \binom{2+2}{2} = \mathbf{6}$

SOS and SDP

- **Example:** Let's try to apply the Theorem to $p(x) = x^4 + 4x^3 + 6x^2 + 4x + 5$
  - From the Theorem we know: $\boldsymbol{p(x)}$ **is SOS** $\Leftrightarrow$ $\exists \boldsymbol{Q} \succcurlyeq \boldsymbol{0}$ **s.t.** $\boldsymbol{p(x) = z(x)^T Q z(x)}$
  - How can we construct the vector of monomials $z(x)$??
  - What is the $\mathbf{deg}(\boldsymbol{p}(\boldsymbol{x}))$ and $\boldsymbol{n}$??

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  - $\deg(p(x)) = 4$ and $n = 1$
  - $z(x) = (1, x, x^2)^T$

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  - How can we construct the vector of monomials $z(x)$??
  - $\deg(p(x)) = 4$ and $n = 1$
  - $z(x) = (1, x, x^2)^T$
  - Then we need to find $Q \succcurlyeq 0$ such that: $\mathrm{p(x)} = \mathrm{z(x)^T Q z(x)} = (1, x, x^2)^T \begin{pmatrix} q_{11} & q_{12} & q_{13} \\ q_{12} & q_{22} & q_{23} \\ q_{13} & q_{23} & q_{33} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix}$

SOS and SDP

- Equivalently: $x^4 + 4x^3 + 6x^2 + 4x + 5 = (1, x, x^2)^T \begin{pmatrix} q_{11} & q_{12} & q_{13} \\ q_{12} & q_{22} & q_{23} \\ q_{13} & q_{23} & q_{33} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix} \Rightarrow \cdots$

$$\ldots \Rightarrow x^4 + 4x^3 + 6x^2 + 4x + 5 = \mathrm{q}_{33}\mathrm{x}^4 + 2\mathrm{q}_{23}\mathrm{x}^3 + (2\mathrm{q}_{13} + \mathrm{q}_{22})\mathrm{x}^2 + 2\mathrm{q}_{12}\mathrm{x} + \mathrm{q}_{11} \Rightarrow$$

$$\Rightarrow \quad \begin{matrix} q_{33} = 1, \\ 2q_{23} = 4, \\ 2q_{13} + q_{22} = 6, \end{matrix} \quad \begin{matrix} 2q_{12} = 4, \\ q_{11} = 5 \end{matrix} \qquad (4)$$

- The **question** we have now is:
  - Can we find a 3×3 PSD matrix $Q \succcurlyeq 0$ that satisfies the system of equations (4) ??

SOS and SDP

- Equivalently: $x^4 + 4x^3 + 6x^2 + 4x + 5 = (1, x, x^2)^T \begin{pmatrix} q_{11} & q_{12} & q_{13} \\ q_{12} & q_{22} & q_{23} \\ q_{13} & q_{23} & q_{33} \end{pmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix} \Rightarrow$

$$\ldots \Rightarrow x^4 + 4x^3 + 6x^2 + 4x + 5 = \mathrm{q}_{33}\mathrm{x}^4 + 2\mathrm{q}_{23}\mathrm{x}^3 + (2\mathrm{q}_{13} + \mathrm{q}_{22})\mathrm{x}^2 + 2\mathrm{q}_{12}\mathrm{x} + \mathrm{q}_{11} \Rightarrow \ldots$$

$$\Rightarrow \begin{array}{c} q_{33} = 1, \\ 2q_{23} = 4, \\ 2q_{13} + q_{22} = 6, \end{array} \quad \begin{array}{c} 2q_{12} = 4, \\ q_{11} = 5 \end{array} \qquad (4)$$

- The **question** we have now is:
  - Can we find a 3×3 PSD matrix $Q \succcurlyeq 0$ that satisfies the system of equations (4) ??

What type of problem is this ??

SOS and SDP

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- The **question** we have now is:
  - Can we find a 3×3 PSD matrix $Q \succcurlyeq 0$ that satisfies the system of equations (4) ??

What type of problem is this ?? → A **feasibility SDP problem** (has only constraints & no objective)

SOS and SDP

- Hence, we have the **feasibility SDP problem**

Any constant is OK

$$\min_{Q \in S^{3\times 3}} \quad 1$$

$$s.t. \quad \begin{array}{c} q_{33} = 1, \\ 2q_{23} = 4, \\ 2q_{13} + q_{22} = 6, \\ \mathrm{Q} \succcurlyeq 0 \end{array} \quad \begin{array}{c} 2q_{12} = 4, \\ q_{11} = 5 \end{array} \qquad (5)$$

- Give this SDP problem to the CVX solver and you will get the PSD matrix: $\mathrm{Q} = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 6 & 2 \\ 0 & 2 & 1 \end{bmatrix} \succcurlyeq 0$

- **Since $Q \succcurlyeq 0$ we know that $p(x) = z(x)^T Q z(x)$ is SOS**

- The **explicit form** of $p(x)$ as SOS is given by **factorizing** $Q = M^T M$, with $\mathrm{M} = \begin{bmatrix} 0 & 2 & 1 \\ \sqrt{2} & \sqrt{2} & 0 \\ \sqrt{3} & 0 & 0 \end{bmatrix}$

SOS and SDP

- Doing the algebra:

$$p(x) = z(x)^T Q z(x) = z(x)^T M^T M z(x) = \big(Mz(x)\big)^T Mz(x) = \big|\big|Mz(x)\big|\big|^2$$

- Compute the vector $Mz(x)$:

$$\mathrm{Mz(x)} = \begin{bmatrix} 0 & 2 & 1 \\ \sqrt{2} & \sqrt{2} & 0 \\ \sqrt{3} & 0 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ x \\ x^2 \end{pmatrix} = \begin{pmatrix} 2x + x^2 \\ \sqrt{2}(1+x) \\ \sqrt{3} \end{pmatrix}$$

- And hence the **explicit form** of $p(x)$ as an SOS is:

$$\mathrm{p(x)} = \big|\big|\mathrm{Mz(x)}\big|\big|^2 = \left|\left|\begin{pmatrix} 2x + x^2 \\ \sqrt{2}(1+x) \\ \sqrt{3} \end{pmatrix}\right|\right|^2 = (2x + x^2)^2 + 2(1+x)^2 + 3$$

SOS and SDP

- Recall the problems (1), (2) and (3)
- We have shown a way to **determine** the constraint: $p(x)$ is SOS using the feasibility SDP problem (5)
- But I need to show that $p(x) - \gamma$ is SOS to solve problem (3) **and** obtain a **lower bound** of (2) or (1)
  - That is, we want to show:

$$\mathbf{p(x)} - \boldsymbol{\gamma} = \boldsymbol{x^4 + 4x^3 + 6x^2 + 4x + 5 - \gamma} \textbf{ is SOS}$$

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ s.t. \; x \in R^n \end{array}\right\} (1)$$

$$\Updownarrow (=)$$

$$\left.\begin{array}{ll} \max\limits_{\gamma} & \gamma \\ s.t. & \boxed{\begin{array}{c} p(x) - \gamma \geq 0, \forall x \in R^n \\ \gamma \in R \end{array}} \end{array}\right\} (2)$$

Ω_2

$$\geq$$

$$\left.\begin{array}{ll} \max\limits_{\gamma} & \gamma \\ s.t. & \boxed{\begin{array}{c} p(x) - \gamma \text{ is SOS} \\ \gamma \in R \end{array}} \end{array}\right\} (3)$$

Ω_3

SOS and SDP

- If we follow the **same process** as before but use $p(x) - \gamma$ instead of $p(x)$ we will end up with the SDP problem:

$$\left.\begin{array}{l} \min\limits_{x} \; p(x) \\ s.t. \; x \in R^n \end{array}\right\} (1)$$

$$\Updownarrow (=)$$

$$\left.\begin{array}{ll} \max\limits_{\gamma} & \gamma \\ s.t. & \boxed{\begin{array}{c} p(x) - \gamma \geq 0, \forall x \in R^n \\ \gamma \in R \end{array}} \end{array}\right\} (2)$$

Ω_2

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Ω_3

$$\left.\begin{array}{ll} \max\limits_{Q \in S^{3\times 3},\, \gamma} & \gamma \\ s.t. & \begin{array}{ll} q_{33} = 1, & 2q_{12} = 4, \\ 2q_{23} = 4, & q_{11} = 5 - \gamma \\ 2q_{13} + q_{22} = 6, & \\ Q \succcurlyeq 0, \quad \gamma \in R & \end{array} \end{array}\right\} (6)$$

Give this to CVX

- The optimal solution of the SDP (6) is a **lower bound** of the global solution of the nonconvex polynomial optimization problem (1)